

intro

July 25, 2026

1 CV5972 - Optimization Techniques for Transportation Engineering

1.0.1 Objectives

- To develop fundamental understanding of optimization theory and techniques pertaining to linear and non-linear programming.
- To formulate and solve optimization problems in transportation engineering using suitable methods.
- To interpret and evaluate solutions derived from optimization models and support informed decision-making in real-world transportation scenarios.

1.0.2 Content

- Module 1. Linear Programming
 - principles of linear programming
 - methods (graphical method, spreadsheet-based solvers, and simplex algorithm)
 - managerial decision-making with sensitivity analysis and duality
 - applications on variants of maximum-flow problem, least-cost path problem, transportation problem, and trans-shipment problem, among others.
- Module 2. Non-Linear Programming - Exact Methods
 - principles of non-linear programming
 - methods (penalty method, barrier method, and interior-point method)
 - managerial decision-making with sensitivity analysis and duality
 - applications on variants of facility location problem and traffic assignment problem, among others.
- Module 3. Non-Linear Programming - Approximate Methods
 - fundamentals of heuristics
 - methods (local search heuristics, evolutionary computation techniques, and swarm intelligence algorithms)
 - managerial decision-making
 - applications on variants of traveling salesman problem and vehicle routing problem, among others.

1.0.3 Textbooks

- Hillier, F. S., & Lieberman, G. J. (2024). Introduction to operations research. (12th edition). McGraw-Hill.

- Winston, W. L. (2022). Operations Research: Applications and Algorithms. (4th edition). Cengage Learning.
- Al-Khateeb, G. (2020). Solved Practical Problems in Transportation Engineering. (1st edition). Taylor & Francis Group.
- Taha, H. A. (2022). Operations Research: An Introduction. (11th edition). Pearson Education India.
- Balakrishnan, N., Render, B., & Stair, R. M. (2013). Managerial decision modelling with spreadsheets. (3rd edition). Prentice Hall Press.

1.0.4 Schedule

S. No.	Topic
01	Linear Programming - Principles
02	Problem Types
03	Linear Programming
04	Graphical Solution Method
05	Spreadsheet Solver Method
06	Python SciPy Method
07	Simplex Method
08	Sensitivity Analysis — Linear Programming
09	Duality — Linear Programming
10	Transportation Engineering Problems - I
11	Transportation Engineering Problems - II
12	Non-Linear Programming - Principles
13	Unconstrained Optimization - Gradient Descent
14	Unconstrained Optimization - Newton's Method
15	Unconstrained Optimization - Quasi-Newton's Method
16	Constrained Optimization - Penalty Method
17	Constrained Optimization - Barrier Method
18	Constrained Optimization - Interior-Point Method
19	Sensitivity Analysis — Non-Linear Programming
20	Duality — Non-Linear Programming
21	Transportation Engineering Problems - III
22	Transportation Engineering Problems - IV
23	Metaheuristics
24	Hill Climb
25	Threshold Accepting
26	Simulated Annealing
27	Tabu Search
28	Iterative Local Search
29	Variable Neighbourhood Search
30	Evolutionary Computation
31	Swarm Intelligence
32	Transportation Engineering Problems - V
33	Transportation Engineering Problems - VI

1.0.5 TimeTable

1.0.6 Grading

Assignments (20%)

S. No.	Content	Marks
1	Lectures 01 - 06	50
2	Lectures 07 - 11	50
3	Lectures 12 - 15	50
4	Lectures 16 - 22	50
5	Lectures 23 - 33	100

Quiz (40%)

S. No.	Content	Marks
1	Lectures 01 - 11	50
2	Lectures 12 - 22	50

End Sem (40%)

Content	Marks
Lectures 01 - 11	25
Lectures 12 - 22	25
Lectures 23 - 33	50